Attendees:
GE: Mike Russo, Todd Shirbak
LTI: Nam Park, Earle Flynn

DT4D / AWS

Notes:
HPC and MlaaS

DT4D/AWS Project Summary

Background

GE Vernova's wind and power businesses are currently spending millions with Quantum Black (a McKinsey-owned firm) to build surrogate modeling platforms—capabilities that replicate what the original DT4D project once did. ARC leadership recognizes this represents both an opportunity and a gap: these are capabilities GRC previously built internally and should be able to develop again.

Project Overview

GE Vernova Research is launching an initiative to rebuild surrogate modeling capabilities internally using Quantum Black's architecture as a reference blueprint. Rather than maintaining or extending Quantum Black's platform (which will be owned by the wind business after their contract ends), we're building our own version from the ground up within ARC's AWS environment. The platform will be packaged as "surrogate in a box"—a shared service offering that other GE Vernova business units can access.

Technology Stack: Kubernetes, S3 storage, Ansys CFX, Java Spring backend, React frontend, with AWS burstable compute for scalability.

Goal

Develop ARC's own internal surrogate modeling capability that:

- Leverage existing LTI DT4D knowledge and team expertise

- Reduces external vendor dependency and associated costs
- Builds AWS cloud expertise within ARC
- Serves as a reusable tool across GE Vernova business units (wind, power, electrification, etc.)

Next Steps

1. **Documentation Review** — Nam, Earl, and Dave Garber to review DT4 D and Quantum Black architecture documentation
2. **Prepare Feedback** — Identify questions, gaps, and improvement opportunities
3. **Internal Follow-up Call** — Tuesday at 9:30am ET; Mike Young will send an email to Michael Russo for confirmation of understanding and more high level kickoff details
4. **Business Case** — Michael Russo presenting to leadership next week for formal approval and funding
5. **Resource Coordination** — Begin with core LTI team; scale resources as needed based on project requirements

Questions:

1. MY - What is the plan for the app stack layer (Java, react?)
2. MY - At this point, do you know when this project might kick off? Nam does have a couple other projects but to me this seems like the best fit for him. I can hold him for a bit but any additional clarity there would be helpful.
3. EF - Are we actually rebuilding from the ground up or extending?